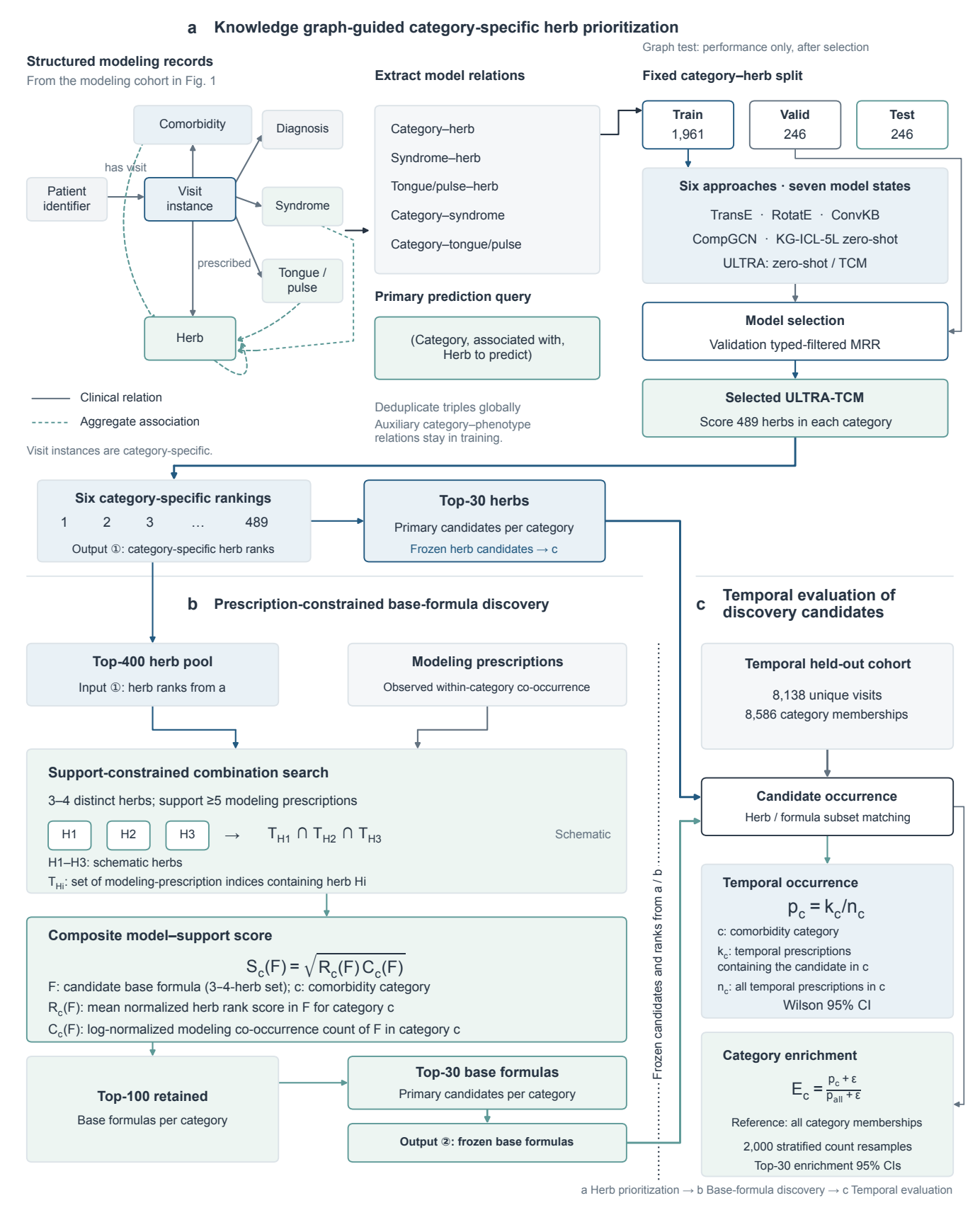

a Knowledge graph-guided category-specific herb prioritization
Structured modeling records
From the modeling cohort in Fig. 1
Comorbidity
Diagnosis
has visit
Patient identifier
Visit instance
Syndrome
prescribed
Tongue / pulse
Herb
Clinical relation
Aggregate association
Visit instances are category-specific.
Extract model relations
Category–herb
Syndrome–herb
Tongue/pulse–herb
Category–syndrome
Category–tongue/pulse
Primary prediction query
(Category, associated with, Herb to predict)
Deduplicate triples globally
Auxiliary category–phenotype relations stay in training.
Graph test: performance only, after selection
Fixed category–herb split
Train 1,961
Valid 246
Test 246
Six approaches · seven model states
TransE · RotatE · ConvKB
CompGCN · KG-ICL-5L zero-shot
ULTRA: zero-shot / TCM
Model selection
Validation typed-filtered MRR
Selected ULTRA-TCM
Score 489 herbs in each category
Six category-specific rankings
1 2 3 … 489
Output ①: category-specific herb ranks
Top-30 herbs
Primary candidates per category
Frozen herb candidates → c
b Prescription-constrained base-formula discovery
Top-400 herb pool
Input ①: herb ranks from a
Modeling prescriptions
Observed within-category co-occurrence
Support-constrained combination search
3–4 distinct herbs; support ≥5 modeling prescriptions
H1 H2 H3 → $T_{H1} \cap T_{H2} \cap T_{H3}$
Schematic
H1–H3: schematic herbs
T_{Hi}: set of modeling-prescription indices containing herb Hi
Composite model–support score
$S_c(F) = \sqrt{R_c(F)\,C_c(F)}$
F: candidate base formula (3–4-herb set); c: comorbidity category
$R_c(F)$: mean normalized herb rank score in F for category c
$C_c(F)$: log-normalized modeling co-occurrence count of F in category c
Top-100 retained
Base formulas per category
Top-30 base formulas
Primary candidates per category
Output ②: frozen base formulas
Frozen candidates and ranks from a / b
c Temporal evaluation of discovery candidates
Temporal held-out cohort
8,138 unique visits
8,586 category memberships
Candidate occurrence
Herb / formula subset matching
Temporal occurrence
$p_c = k_c/n_c$
c: comorbidity category
k_c: temporal prescriptions containing the candidate in c
n_c: all temporal prescriptions in c
Wilson 95% CI
Category enrichment
$E_c = \frac{p_c + \varepsilon}{p_{all} + \varepsilon}$
Reference: all category memberships
2,000 stratified count resamples
Top-30 enrichment 95% CIs
a Herb prioritization → b Base-formula discovery → c Temporal evaluation